

Boom Lift-mounted Robot

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***Manufacturing and Aviation
Automation Lab***

Standard content

- First-level body text uses the same restrained Arial-style hierarchy.
 - Second-level points are smaller and indented.
 - Keep important evidence close to the point it supports.
- Use **red emphasis** sparingly.

Important formula

$$q_k = \frac{\varphi_k^T F}{\varphi_k^T K \varphi_k}$$

Hardware – Sensor

IMU	Category	Frequency	Price
HWU0703	HWU1703 (RS485)	100 Hz	2,371
IMU5731	—	500 Hz	4,500
IMU5735	—	400 Hz	4,800
AHRS731	—	100 Hz	5,800

Two-column layout

Research question

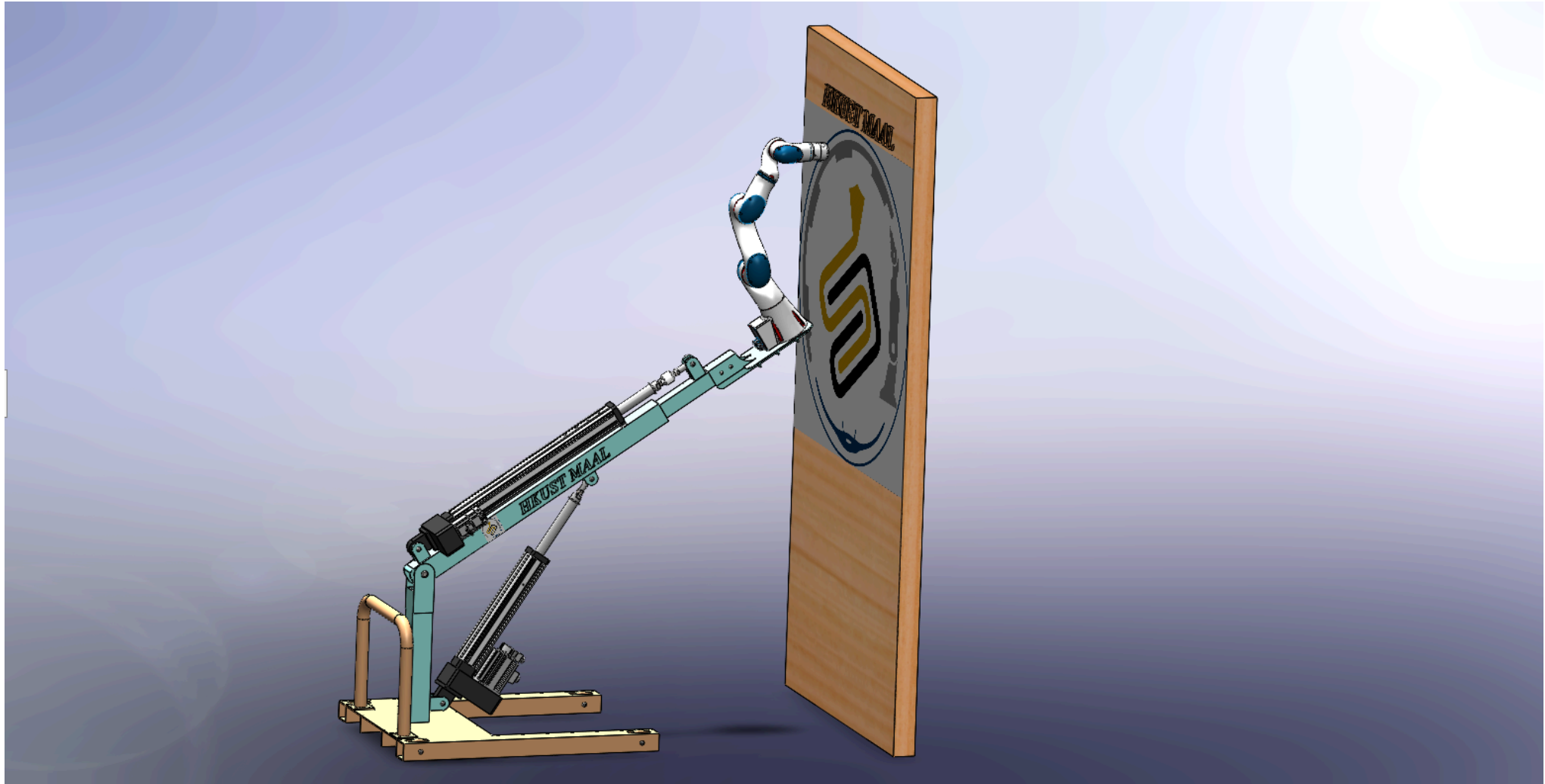
- What is the dominant mechanism?
- Which evidence supports the claim?
- What remains uncertain?

Key takeaway

State the result directly, then show the evidence beneath it.

Need to improve quality, flexibility, and productivity

Hardware – Design



Process and relation illustration



Input

Experimental data and physical constraints

Output

Validated model and actionable conclusion

References with pictures

Replace with figure 1

Method or platform name

(Author et al., Year)

Replace with figure 2

Block diagram or result

(Author et al., Year)

Summary slide example

- Energy-efficient machining achieved via hybrid feed drive and control allocation
- Combining **mechatronics** and **optimal control allocation** improves energy efficiency without sacrificing accuracy
 - About 80% energy reduction from LMD to HFD
 - A further 50% reduction with optimal allocation
- Smart control uses process data to improve quality and sustainability
- Key techniques
 - Reconfigurable mechanisms
 - Input redundancy and control allocation
 - Intelligent control informed by process data